F18CF – Linear Algebra

Tutorial Sheet 3 – Solutions

1. (a) We need to determine whether there exist $a, b \in \mathbb{R}$ such that

$$a(x^2 + x + 1) + b(x^2 - 3x - 1) = x^2 + 3x + 2.$$

By rearranging this equation, we get the equation:

$$(a + b)x^2 + (a - 3b)x + (a - b) = x^2 + 3x + 2.$$

Since two polynomial functions are equal if and only if they have the same coefficient, we have to solve the system:

$$\begin{cases} a + b &= 1 \\ a - 3b &= 3 \\ a - b &= 2 \end{cases}$$

which we can solve by Gaussian elimination, or directly, and we find a unique solution: $a = \frac{3}{2}$ and $b = -\frac{1}{2}$.

We thus have

$$\frac{3}{2}(x^2 + x + 1) - \frac{1}{2}(x^2 - 3x - 1) = x^2 + 3x + 2.$$

(b) We need to determine whether there exist $a, b \in \mathbb{R}$ such that

$$a(x^2 + x + 1) + b(x^2 - 3x - 1) = x^2.$$

By rearranging this equation, we get the equation:

$$(a + b)x^2 + (a - 3b)x + (a - b) = x^2.$$

Since two polynomial functions are equal if and only if they have the same coefficient, we have to solve the system:

$$\begin{cases} a + b &= 1 \\ a - 3b &= 0 \\ a - b &= 0 \end{cases}$$

which we can solve by Gaussian elimination:

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -3 & 0 \\ 1 & -1 & 0 \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 1 \end{array}\right)$$

Since this system has a forbidden row, it is inconsistent. Thus, x^2 is not a linear combination of P_1 and P_2.

2. (a) Assume that there exists a linear combination of the form $\lambda f(x) + \mu g(x) = 0$ for every $x \in \mathbb{R}$. For $x = 0$, we get $\lambda = 0$, and for $x = \pi/2$, we get $\mu = 0$. Thus, $\lambda = \mu = 0$, and f, g are linearly independent.

   (b) Assume that there exists a linear combination of the form $\lambda f(x) + \mu g(x) = \cos(2x)$ for every $x \in \mathbb{R}$. For $x = 0$, we get $\lambda = 1$, and for $x = \pi$, we get $-\lambda = 1$, which is a contradiction. Thus, $\cos(2x)$ is not a linear combination of f and g.

3. (a) We have

$$x^3 = x^3 + 0x^2 + 0x + 0 \quad \rightsquigarrow \quad [P_1]_{\mathcal{B}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$x^2(x-1) = x^3 - x^2 + 0x + 0 \quad \rightsquigarrow \quad [P_1]_{\mathcal{B}} = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}.$$

$$x(x-1)^2 = x^3 - 2x^2 + x + 0 \quad \rightsquigarrow \quad [P_1]_{\mathcal{B}} = \begin{pmatrix} 0 \\ 1 \\ -2 \\ 1 \end{pmatrix}.$$

$$(x-1)^3 = x^3 - 3x^2 + 3x - 1 \quad \rightsquigarrow \quad [P_1]_{\mathcal{B}} = \begin{pmatrix} -1 \\ 3 \\ -3 \\ 1 \end{pmatrix}.$$

   (b) Using results from the course, we have:

$$P_1, \ldots, P_4 \text{ is a basis of } \mathcal{P}_3(\mathbb{R}) \Leftrightarrow [P_1]_{\mathcal{B}}, \ldots, [P_4]_{\mathcal{B}} \text{ is a basis of } \mathbb{R}^4 \Leftrightarrow \det\left([P_1]_{\mathcal{B}} \middle| \cdots \middle| [P_4]_{\mathcal{B}}\right) \neq 0.$$

   We have

$$\det\left([P_1]_{\mathcal{B}} \middle| \cdots \middle| [P_4]_{\mathcal{B}}\right) = \det\begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & -1 & -2 & 3 \\ 1 & 1 & 1 & 1 \end{pmatrix} = -\det\begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 3 \\ -1 & -2 & 3 \end{pmatrix} =$$

$$= \det\begin{pmatrix} 0 & -1 \\ 1 & 3 \end{pmatrix} = 1.$$

   Thus, $P_1, \ldots, P_4$ is a basis of $\mathcal{P}_3(\mathbb{R})$.

4. (a) From the definition of L_j we see that $L_j(x_i) = 0$ if $i \neq j$.

   By evaluating the equation $\alpha_1 L_1(x) + \ldots + \alpha_n L_n(x) = 0$ at $x = x_i$, we get

$$0 + \ldots + 0 + \alpha_i \times \underbrace{(x_i - x_1) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)}_{\neq 0} + 0 + \ldots + 0 = 0.$$

Since the x_i are all distinct, we get $\alpha_i = 0$.

We thus have $\alpha_1 L_1 + \ldots + \alpha_n L_n = 0 \Rightarrow \alpha_1 = \ldots = \alpha_n = 0$, hence the family $L_1, \ldots, L_n$ is linearly independent.

(b) Each polynomial L_i has degree $n-1$, hence they belong to $\mathcal{P}_{n-1}(\mathbb{R})$, which is a vector space of dimension n. Since $L_1, \ldots, L_n$ is a family of n linearly independent vectors in a space of dimension n, we know from the course that it is also a basis, hence $L_1, \ldots, L_n$ is a basis of $\mathcal{P}_{n-1}(\mathbb{R})$.

(c) We want to find a polynomial function $P \in \mathcal{P}_{n-1}(\mathbb{R})$ such that $P(x_i) = y_i$ for every $i = 1, \ldots, n$. Since $L_1, \ldots, L_n$ is a basis of $\mathcal{P}_{n-1}(\mathbb{R})$, we write such a polynomial P as a linear combination of the L_i's, that is,

$$P = \alpha_1 L_1 + \ldots + \alpha_n L_n.$$

Evaluating at $x = x_i$, we get $y_i = P(x_i) = \alpha_i (x_i - x_1) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)$, hence

$$\alpha_i = \frac{y_i}{(x_i - x_1) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)}.$$

Thus, there is exactly one choice of coefficients $\alpha_1, \ldots, \alpha_n$ that satisfies the hypotheses, which gives a unique polynomial in $\mathcal{P}_{n-1}(\mathbb{R})$ satisfying the hypotheses.

(d) We have the polynomials $L_1(x) = (x-2)(x-3)$, $L_2(x) = (x-1)(x-3)$, and $L_3(x) = (x-1)(x-2)$. The interpolating is given by $P(x) = \alpha_1 L_1(x) + \alpha_2 L_2(x) + \alpha_3 L_3(x)$, with

$$\alpha_1 = \frac{1}{(1-2)(1-3)} = \frac{1}{2}, \quad \alpha_2 = \frac{8}{(2-1)(2-3)} = -8, \quad \alpha_3 = \frac{27}{(3-1)(3-2)} = \frac{27}{2}.$$

We thus have

$$P(x) = \frac{1}{2}(x-2)(x-3) - 8(x-1)(x-3) + \frac{27}{2}(x-1)(x-2) = 6x^2 - 11x + 6.$$

5. (a) Let $a_1, \ldots, a_k \in \mathbb{R}$ such that $a_1 \delta_1 + \cdots + a_k \delta_k = \vec{0}$. This is equivalent to

$$a_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix} + \cdots + a_k \begin{pmatrix} \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \end{pmatrix} = \begin{pmatrix} a_1 \\ \vdots \\ a_k \\ 0 \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}.$$

Since two sequences are equal if and only if all their terms agree, we get $a_1 = \cdots = a_k = 0$, hence the sequences $\delta_1, \ldots, \delta_k$ are linearly independent.

(b) Since $\delta_1, \ldots, \delta_k$ are linearly independent from (a), it follows that $\dim \mathbb{R}^{\mathbb{N}} \geq k$. As this is true for every $k \geq 1$, we get $\dim \mathbb{R}^{\mathbb{N}} = \infty$.

6. (a) Evaluating the equation $ae^x + be^{2x} + ce^{4x} = 0$ at $x = 0$ yields $a + b + c = 0$.

   (b) Since the two functions $ae^x + be^{2x} + ce^{4x}$ and 0 are equal by assumption, their derivatives are also equal. Differentiating the equation $ae^x + be^{2x} + ce^{4x} = 0$ thus yields the equation $ae^x + 2be^{2x} + 4ce^{4x} = 0$ for every $x \in \mathbb{R}$. Evaluating at $x = 0$, we get $a + 2b + 4c = 0$.

   Differentiating the equation $ae^x + 2be^{2x} + 4ce^{4x} = 0$ (which holds for every $x \in \mathbb{R}$) yields the equation $ae^x + 4be^{2x} + 16ce^{4x} = 0$. Evaluating at $x = 0$, we get $a + 4b + 16c = 0$.

   (c) We thus have the following system of linear equations:

$$\begin{cases} x + y + z &= 0 \\ x + 2y + 4z &= 0 \\ x + 4y + 16z &= 0 \end{cases}$$

   We apply Gaussian elimination

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 0 \\ 1 & 4 & 16 & 0 \end{array}\right) \overset{\cdots}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 6 & 0 \end{array}\right)$$

   Every variable being a pivot variable, the system has a unique solution, namely $a = b = c = 0$.

   We thus have

$$ae^x + be^{2x} + ce^{4x} = 0 \text{ for every } x \in \mathbb{R} \quad \Longrightarrow \quad a = b = c = 0.$$

   Therefore, the functions e^x, e^{2x}, e^{4x} are linearly independent.

7. (a) Let $\lambda_1, \ldots, \lambda_k \in \mathbb{R}$ such that

$$\lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k = \vec{0}.$$

   We need to show that $\lambda_1 = \cdots = \lambda_k = 0$. By applying the coordinatisation map T to both sides of this equation and using the properties (linearity) of T, we get

$$T\Big(\lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k\Big) = \lambda_1 T(\vec{v}_1) + \cdots + \lambda_k T(\vec{v}_k) = T(\vec{0}) = \vec{0}.$$

   But since $T(\vec{v}_1), \ldots, T(\vec{v}_k)$ is linearly independent in $\mathbb{R}^n$, it follows that $\lambda_1 = \cdots = \lambda_k = 0$. Thus, $\vec{v}_1, \ldots, \vec{v}_k$ is linearly independent in V.

   (b) Let $\vec{v} \in V$. Let us look at its image $T(\vec{v}) \in \mathbb{R}^n$. Since $T(\vec{v}_1), \ldots, T(\vec{v}_k)$ spans $\mathbb{R}^n$, there exists $\lambda_1, \ldots, \lambda_k \in \mathbb{R}$ such that

$$T(\vec{v}) = \lambda_1 T(\vec{v}_1) + \cdots + \lambda_k T(\vec{v}_k).$$

   Using the linearity of T, this can be rewritten as

$$T(\vec{v}) = T(\lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k).$$

But since T is a bijection, we have

$$T(x) = T(y) \iff x = y.$$

Thus, we get that $\vec{v} = \lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k$. As every vector of V can be written as linear combination of $\vec{v}_1, \ldots, \vec{v}_k$, it follows that $\vec{v}_1, \ldots, \vec{v}_k$ spans V.